

## Objective

To develop an AI-based spam classification system that uses **BERT (Bidirectional Encoder Representations from Transformers)** to classify text messages or emails as **Spam** or **Not Spam**. The goal was to fine-tune a pre-trained BERT model on a labeled dataset and achieve high accuracy with real-world usability.

## Tools & Technologies

| Category             | Tools Used                                        |
|----------------------|---------------------------------------------------|
| Programming Language | Python                                            |
| Frameworks           | PyTorch, Hugging Face Transformers                |
| Dataset              | SMS Spam Collection Dataset                       |
| Libraries            | transformers, datasets, torch, scikit-learn, tqdm |
| Platform             | Google Colab                                      |
| Hardware Used        | CPU (runtime: ~1.5 hours for training)            |

## Dataset Details

- **Dataset Source:** Hugging Face sms\_spam dataset
- **Total Samples:** 5,574 messages
- **Train-Test Split:** 80% training (4,459), 20% testing (1,115)
- **Classes:**
  - **0 → Not Spam (Ham)**
  - **1 → Spam**

## Data Preprocessing Steps

1. **Text Cleaning:**
  - Removed URLs, special characters, and excessive spaces.

- Converted text to lowercase.

## 2. Tokenization:

- Used BertTokenizer with max\_length=128, truncation, and padding.

## 3. Data Loader Preparation:

- Converted to PyTorch DataLoader for efficient batching.

### Model Setup

| Component        | Description                   |
|------------------|-------------------------------|
| Base Model       | bert-base-uncased             |
| Architecture     | BertForSequenceClassification |
| Number of Labels | 2 (Spam / Not Spam)           |
| Optimizer        | AdamW (learning rate = 2e-5)  |
| Batch Size       | 8                             |
| Epochs           | 1                             |
| Loss Function    | Cross-Entropy Loss            |

### Training Summary

| Metric          | Value                  |
|-----------------|------------------------|
| Epochs          | 1                      |
| Average Loss    | <b>0.0716</b>          |
| Training Time   | ~1 hr 30 mins (on CPU) |
| Evaluation Time | ~6 mins                |

### Evaluation Results

| Metric           | Score  |
|------------------|--------|
| Accuracy         | 98.92% |
| F1 Score         | 0.9583 |
| Precision (Spam) | 0.99   |
| Recall (Spam)    | 0.93   |

#### Confusion Matrix

|                  | Predicted: Not Spam | Predicted: Spam |
|------------------|---------------------|-----------------|
| Actual: Not Spam | 965                 | 1               |
| Actual: Spam     | 11                  | 138             |

#### Demo Predictions

| Input Message                                      | Predicted Output |
|----------------------------------------------------|------------------|
| "Congratulations! You won a free vacation trip!"   | Not Spam         |
| "Hey, are we still meeting at 5 PM?"               | Not Spam         |
| "URGENT: Claim your \$1000 Walmart gift card now!" | Spam             |
| "Please send the updated project document."        | Not Spam         |

*(Note: Model achieved near-perfect discrimination in test demos.)*

#### Model Storage

- **Saved Path:** /content/spam\_bert\_model
  - Contains:
    - pytorch\_model.bin (fine-tuned weights)
    - config.json
    - vocab.txt

You can copy it to Google Drive using:

```
from google.colab import drive
```

```
drive.mount('/content/drive')
```

```
!cp -r /content/spam_bert_model /content/drive/MyDrive/
```

## Trello Checklist Completion

| Task | Status |
|------|--------|
|------|--------|

|                      |  |
|----------------------|--|
| Install dependencies |  |
|----------------------|--|

|                           |  |
|---------------------------|--|
| Load & preprocess dataset |  |
|---------------------------|--|

|                      |  |
|----------------------|--|
| Fine-tune BERT model |  |
|----------------------|--|

|                            |  |
|----------------------------|--|
| Evaluate with accuracy, F1 |  |
|----------------------------|--|

|                           |  |
|---------------------------|--|
| Generate confusion matrix |  |
|---------------------------|--|

|                         |  |
|-------------------------|--|
| Test with demo messages |  |
|-------------------------|--|

|                          |  |
|--------------------------|--|
| Save model & log results |  |
|--------------------------|--|

## Conclusion

This project successfully fine-tuned **BERT** for spam detection on SMS messages, achieving a **98.9% accuracy** on the test dataset.

The model effectively distinguishes between spam and non-spam texts and can be deployed as a real-time classifier using frameworks like **Flask**, **Gradio**, or **FastAPI**.

Even with CPU-only training, the system demonstrated **excellent generalization and performance**.

## Future Enhancements

- Train for more epochs on GPU for slightly higher recall.
- Deploy as a **web interface (Gradio)** for interactive testing.

- Extend to **email spam detection** using the Enron dataset.
- Optimize with **DistilBERT** for faster real-time inference.